

Comparing properties

Substances can be described and identified in terms of their properties. Chemists compare the properties of materials to find similarities and differences between them. Then they can start to investigate why these similarities and differences exist.

Substance	Floats in water?	Color	Transparency	Luster	Solubility	Conductivity	Texture
copper	no	red	opaque	shiny	in acid	conductor	smooth
natural chalk	no	white	opaque	dull	in acid	insulator	powdery
pencil lead (graphite)	no	black	opaque	shiny	no	conductor	slippery
pine wood	yes	brown	opaque	dull	only in special solvents	insulator	fibrous
salt crystals	no	white	translucent	shiny	in water	insulator when solid	gritty
glass	no	various	varies	shiny	only in special solvents	insulator	smooth
talc	no	various	opaque	waxy	in acid	insulator	greasy
diamond	no	various	transparent	sparkling when cut	no	insulator	smooth

Hardness

The hardness of a substance is normally measured on the Mohs scale, named after its inventor Friedrich Mohs. The scale is based on ten “guide” minerals, which all occur naturally in rocks. The hardness of a substance is measured in comparison with these guides. A material is harder than another when it can leave a scratch on it. For example, a piece of ordinary glass can scratch apatite but not orthoclase, and so its hardness is somewhere between 5 and 6.

▷ Mohs scale

The Mohs scale is only a comparative measure of hardness. In reality, a diamond is not ten times harder than talc. However, the Mohs scale is the preferred measure because it gives meaningful results using a quick and simple method.

1	talc	
2	gypsum	
3	calcite	
4	fluorite	
5	apatite	
6	orthoclase	
7	quartz	
8	topaz	
9	corundum	
10	diamond	

Chemical properties

A substance can be described in terms of its chemical properties. It could be an element (a pure substance that cannot be reduced into simpler constituents), a compound (a combination of two or more elements), or be described as a metal, nonmetal, or semimetal. Chemists also look at a substance’s chemical behavior, cataloguing its reactions and analyzing the products. A full set of properties—chemical and physical—can belong only to one substance.

▷ Reactivity series

Every element has a certain reactivity, which is part of its chemical behavior. Common metals are often ordered by how reactive they are. This is called the reactivity series. Metals at the top are most reactive. Potassium is so reactive that it is rarely found on its own. If two metals are competing to bond with another element, the one higher up the scale would win.

most reactive		
19	K	potassium
11	Na	sodium
20	Ca	calcium
12	Mg	magnesium
13	Al	aluminum
30	Zn	zinc
26	Fe	iron
50	Sn	tin
82	Pb	lead
29	Cu	copper
47	Ag	silver
79	Au	gold
78	Pt	platinum
least reactive		